

1] What are the basic components of SQL syntax?

Ans:

Basic Components of SQL Syntax

1. Statements:

SQL commands used to perform specific tasks. Examples include SELECT, INSERT, UPDATE, DELETE, CREATE, and DROP.

2. Clauses:

Keywords that define the conditions or modify the behaviour of statements. Examples include WHERE, ORDER BY, GROUP BY, and HAVING.

3. Expressions:

Used to produce values, such as calculations or comparisons. Examples: age > 18 or salary * 1.1.

4. Predicates:

Conditions that evaluate to true, false, or unknown. Used in statements like WHERE or JOIN. Example: age > 18 AND city = 'New York'.

5. Data Types:

Defines the type of data stored in a column (e.g., INT, VARCHAR, DATE).

6. Operators:

Symbols or keywords used for calculations, comparisons, or logical operations. Examples: =, >, <, AND, OR, LIKE.

7. Functions:

Predefined operations to manipulate or retrieve data. Examples: COUNT (), SUM (), AVG (), UPPER ().

8. Comments:

Single-line comments: -- This is a comment.

Multi-line comments: /* This is a comment */.

9. Semicolon (;):

Used to terminate SQL statements. It ensures commands are executed properly.